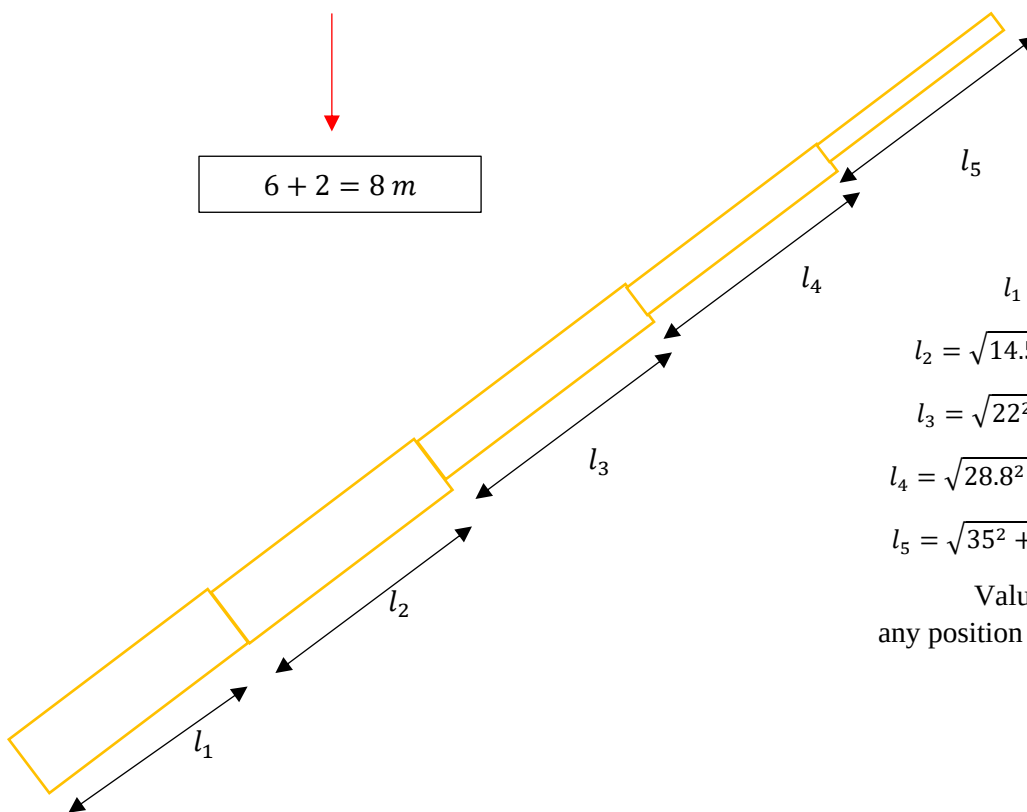
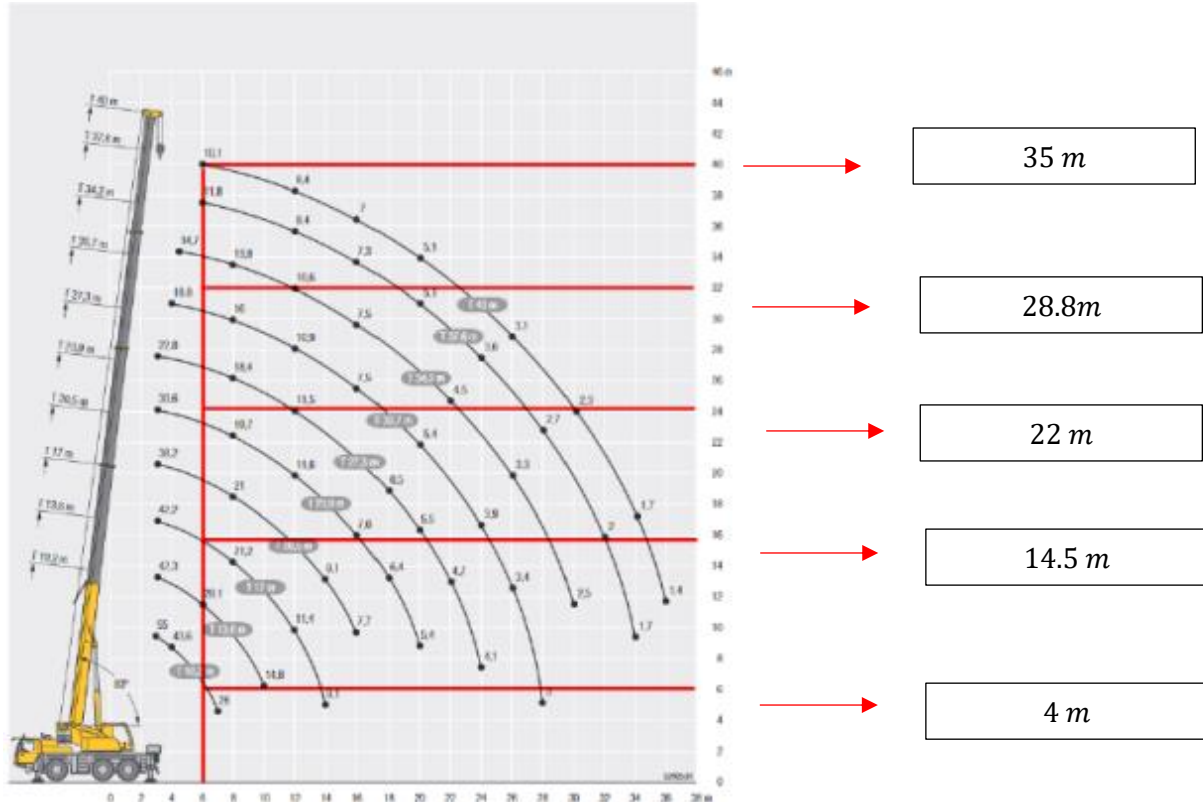


INDRODUCION

In this study, a code has been calculated and written to find the internal forces and moments encountered by the boom of a construction machine when it is under the determined position and weight.

1. Calculating the coordinates and lengths of the booms



$$l_1 = \sqrt{4^2 + 8^2} m = 8.94 m \approx 9 m$$

$$l_2 = \sqrt{14.5^2 + 8^2} = 16.56 - 9 = 7.56 m \approx 7.5 m$$

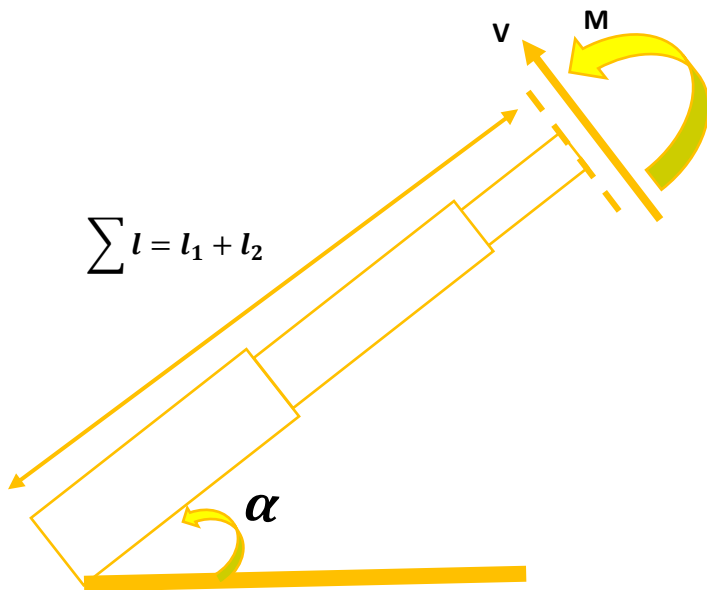
$$l_3 = \sqrt{22^2 + 8^2} = 23.40 - 16.50 = 6.9 m \approx 7 m$$

$$l_4 = \sqrt{28.8^2 + 8^2} = 29.89 - 23.40 = 6.49 m \approx 6.5 m$$

$$l_5 = \sqrt{35^2 + 8^2} = 35.9026 - 29.89 = 6.01 m \approx 6 m$$

Values have been calculated for lengths at any position at given table.

2. Internal Forces for any Position



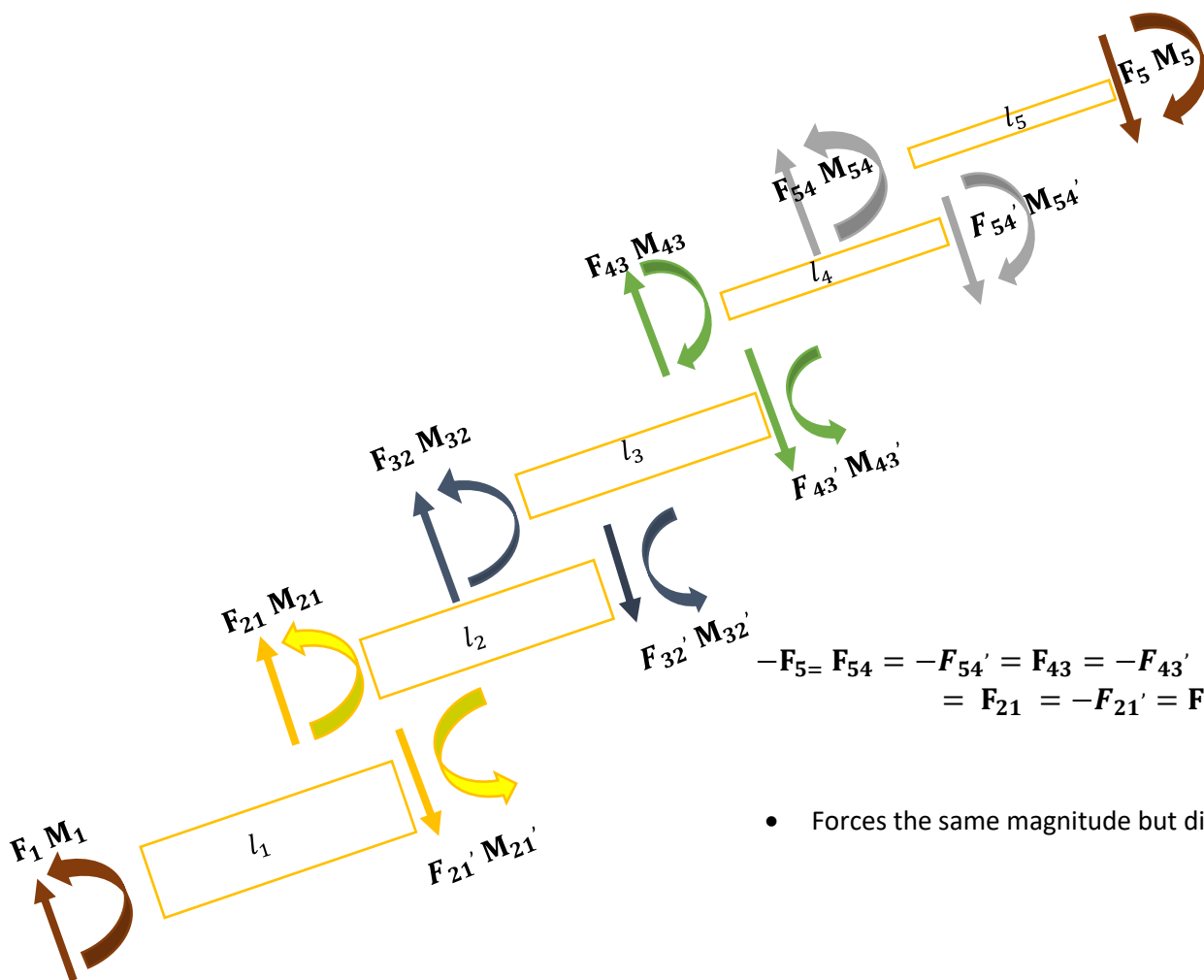
- $\sum F = 0$ (in y direction)

$$F = V$$

for lifting; $F = mg$

- $\sum M = 0$ (CCW)

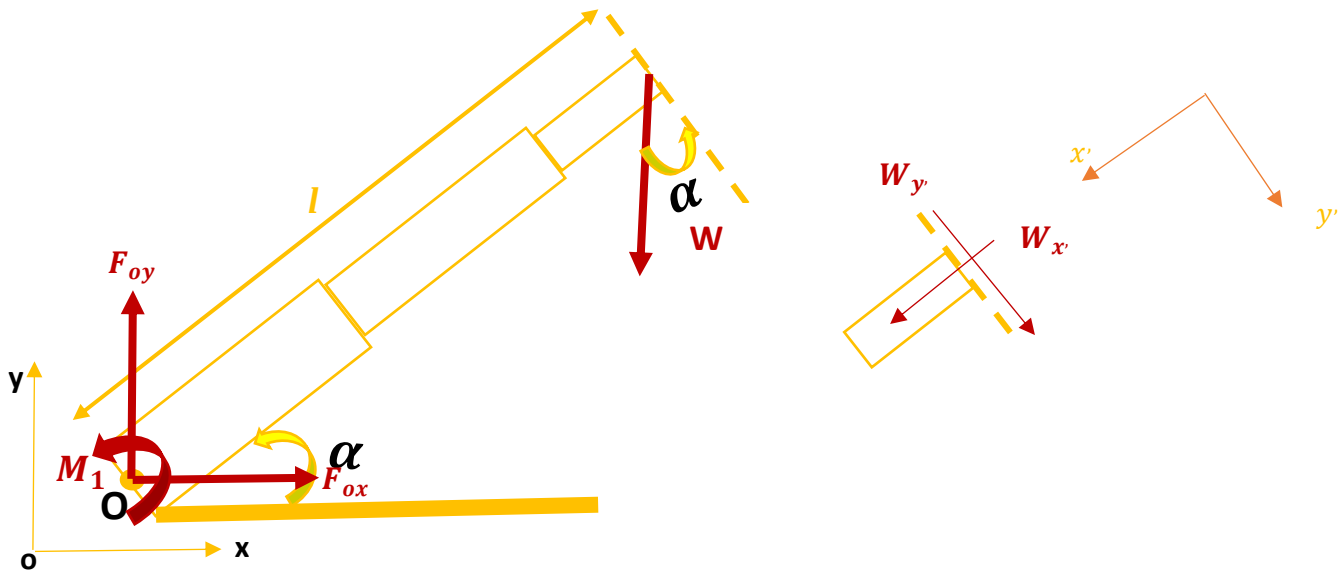
$$M = F \times \cos \alpha \times \sum l$$



$$-F_5 = F_{54} = -F_{54}' = F_{43} = -F_{43}' = F_{32} = -F_{32}' = F_{21} = -F_{21}' = F_1$$

- Forces the same magnitude but different directions.

3. Calculation for any lifting position



- $\sum F_y = 0$ (in y direction)

$$F_{oy} = W \times 9.81 \quad ; \text{lifting force}$$

$$W_{x'} = W \times 9.81 \times \sin \alpha$$

$$W_{y'} = W \times 9.81 \times \cos \alpha$$

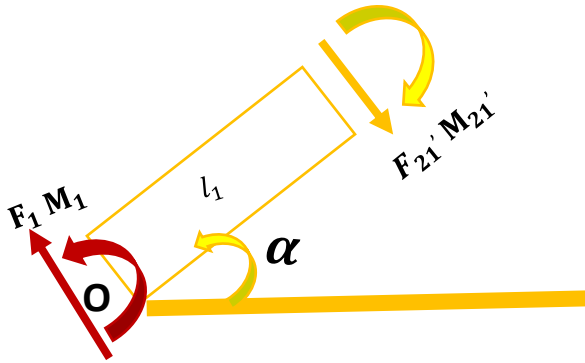
- $\sum F_x = 0$ (in x direction)

$$F_{ox} = 0$$

- $\sum M_o = 0$ (CCW is positive direction)

$$M_1 = W \times 9.81 \times \cos \alpha \times l$$

3.1. Internal Boom Force Calculation

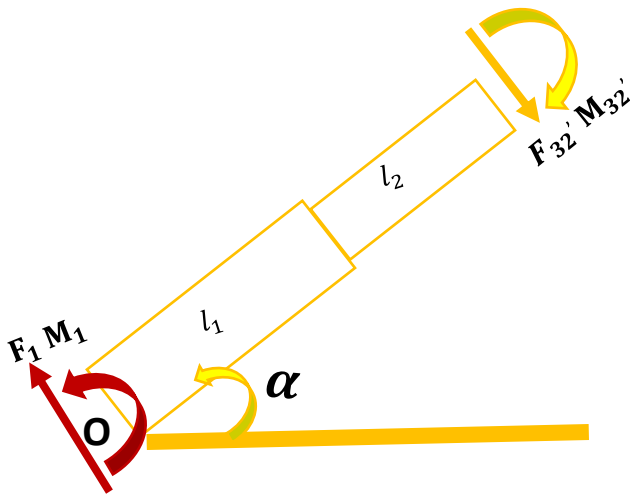


- $\sum M_o = 0$

$$M_1 - M_{21}' - F_{21}' \times l_1 = 0$$

- $\sum F = 0$

$$F_{21}' = F_1$$



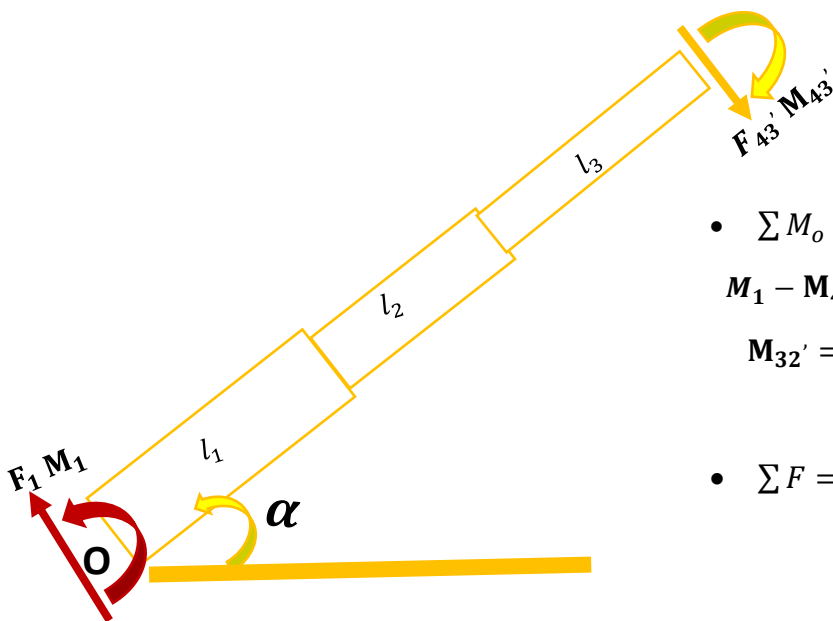
- $\sum M_o = 0$

$$M_1 - M_{32}' - F_{32}' \times (l_1 + l_2) = 0$$

$$M_{32}' = M_1 - F_{32}' \times (l_1 + l_2)$$

- $\sum F = 0$

$$F_{32}' = F_1$$



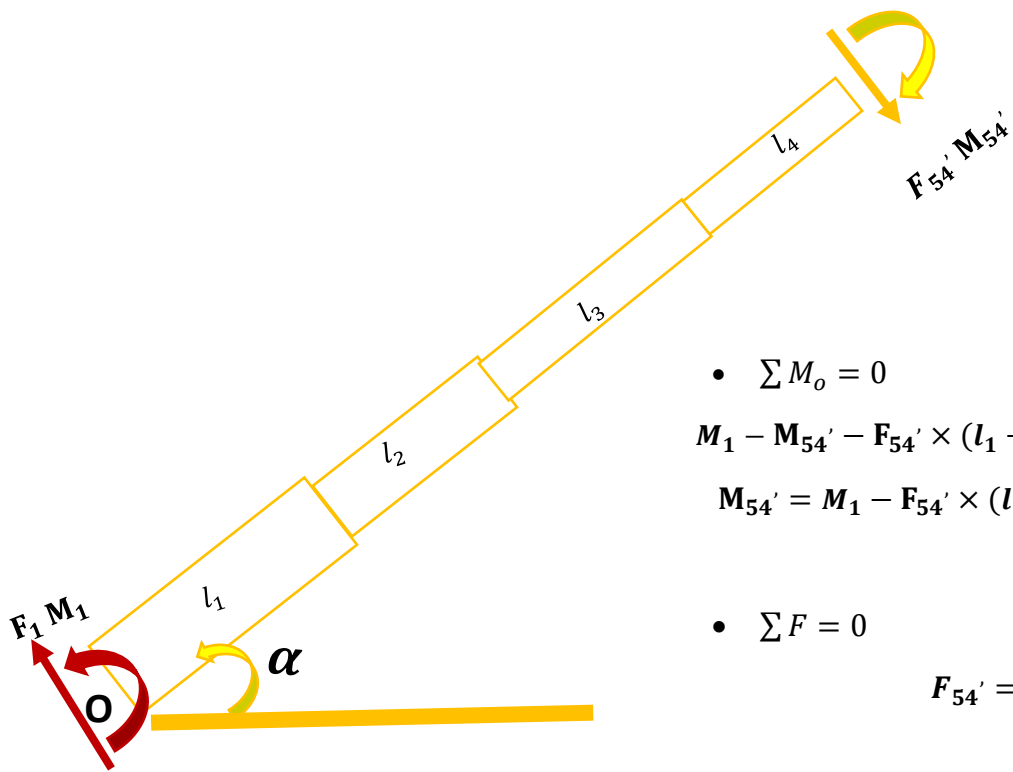
- $\sum M_o = 0$

$$M_1 - M_{43}' - F_{43}' \times (l_1 + l_2 + l_3) = 0$$

$$M_{32}' = M_1 - F_{32}' \times (l_1 + l_2 + l_3)$$

- $\sum F = 0$

$$F_{43}' = F_1$$



- $\sum M_o = 0$

$$M_1 - M_{54}' - F_{54}' \times (l_1 + l_2 + l_3 + l_4) = 0$$

$$M_{54}' = M_1 - F_{54}' \times (l_1 + l_2 + l_3 + l_4)$$

- $\sum F = 0$

$$F_{54}' = F_1$$

Table

POINT(Tonnes)	X	Y	HYPOTENUS	ARCTAN ANGLES(Degree)	FORCE(kN)	MOMENT(Origin-kN.m)
26,0	9	0,3	9,005	1,9092	254,9184	2295,5
43,6	6	5,0	7,8102	39,8056	328,5805	2566,3
55,0	5	5,3	7,2863	46,6683	370,2502	2697,8
14,8	12	2,0	12,1655	9,4623	143,2126	1742,3
29,1	8	7,3	10,8301	42,3805	210,8732	2283,8
42,3	5	9,2	10,4709	61,4769	198,1504	2074,8
9,1	16	1,0	16,0312	3,5763	89,0972	1428,3
11,4	14	6,0	15,2315	23,1986	102,7917	1565,7
21,2	10	10,2	14,2843	45,5673	145,5953	2079,7
42,2	5	13,0	13,9284	68,9625	148,6109	2069,9
7,7	18	5,8	18,9114	17,8601	71,8967	1359,7
9,1	16	9,2	18,4564	29,8989	77,3896	1428,3
21,0	10	14,3	17,4496	55,0349	118,0597	2060,1
38,2	5,4	16,4	17,2662	71,7749	117,2008	2023,6
5,4	22	5,0	22,561	12,8043	51,6567	1165,4
6,4	20	9,4	22,0989	25,1735	56,821	1255,7
7,6	18	12,0	21,6333	33,6901	62,0343	1342
11,6	14	15,8	21,1102	48,4566	75,468	1593,1
19,7	10	18,4	20,9418	61,4769	92,2828	1932,6
30,6	5	20,2	20,8096	76,0973	72,1268	1500,9
4,1	26	3,3	26,2086	7,2335	39,9009	1045,7
4,7	24	9,0	25,632	20,556	43,1713	1106,6
5,5	22	12,2	25,1563	29,0104	47,1854	1187
6,5	20	14,8	24,8805	36,5014	51,257	1275,3
11,5	14	20,0	24,4131	55,008	64,6952	1579,4
18,4	10	22,2	24,3483	65,7508	74,1341	1805
22,8	5	23,2	23,7327	77,8378	47,1224	1118,3
3,0	30	1,1	30,0202	2,0999	29,4102	882,9
3,4	28	8,3	29,2043	16,5113	31,9786	933,9
3,9	26	12,8	28,98	26,2114	34,3249	994,7
5,4	22	17,8	28,2991	38,976	41,1825	1165,4
7,5	18	21,3	27,8871	49,7998	47,4897	1324,3
10,9	14	24,0	27,7849	59,7436	53,8784	1497
16,0	10	26,0	27,8568	68,9625	56,3454	1569,6
18,8	6	27,0	27,6586	77,4712	40,0081	1106,6
2,5	32	7,7	32,9134	13,5296	23,8444	784,8
3,3	28	15,8	32,1503	29,4354	28,194	906,4
4,5	24	20,7	31,6937	40,7778	33,4287	1059,5
7,5	18	25,2	30,9684	54,4623	42,7646	1324,4
10,6	14	28,0	31,305	63,4349	46,504	1455,8
13,8	10	29,2	30,8649	71,0954	43,8615	1353,8
14,7	6,3	30,2	30,8501	78,2166	29,449	908,5
1,7	36	5,3	36,388	8,375	16,4992	600,4
2,0	34	11,8	35,9894	19,1398	18,5354	667,1
2,7	30	19,0	35,5106	32,3474	22,3767	794,6
3,6	26	23,2	34,8459	41,7428	26,3507	918,2
5,1	22	27,0	34,8281	50,8263	31,6032	1100,7
7,3	18	29,4	34,4726	58,5232	37,393	1289
9,4	14	31,3	34,2883	65,9018	37,6512	1291
11,8	8	33,2	34,1503	76,4521	27,1173	926,1
1,4	38	7,6	38,7525	11,3099	13,4673	521,9
1,7	36	13,3	38,0433	18,8636	15,7813	600,4
2,3	32	20,0	37,7359	32,0054	19,1334	722
3,1	28	24,8	37,4037	41,5318	22,7653	851,5
5,1	22	30,0	37,2022	53,7462	29,5865	1100,7
7,0	18	32,5	37,1517	61,0203	33,2706	1236,1
8,4	14,0	34,0	36,9546	67,7379	31,2182	1153,7
10,1	8,0	36,0	36,8782	77,4712	21,4937	792,6

Table 1: Calculated Data in MATLAB

4. Matlab Code

```
>> L1=9;
L2=7.5;
L3=7;
L4=6.5;
L5=6;
L_Total=L1+L2+L3+L4+L5;
L=input('Enter Length of The Force Position in Meter\n');
W=input('Enter Value of W in Tonnes\n');
Alpha=input('Enter Value of Angle Alpha\n');
>> if L<=L_Total & L>=(L_Total-L5);
V0=W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2= W.*1000.*9.81.*cosd(Alpha);
V3= W.*1000.*9.81.*cosd(Alpha);
V4= W.*1000.*9.81.*cosd(Alpha);
M0=V0.*L;
M1=M0-(V0.*L1);
M2=M0-((L1+L2).*V0);
M3=M0-((L1+L2+L3).*V0);
M4=M0-((L1+L2+L3+L4).*V0);
else
if L<(L_Total-L5) & L>=(L_Total-L5-L4);
V0= W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2= W.*1000.*9.81.*cosd(Alpha);
V3= W.*1000.*9.81.*cosd(Alpha);
V4=0;
M0=V0.*L;
M1=M0-(V0.*L1);
M2=M0-((L1+L2).*V0);
M3=M0-((L1+L2+L3).*V0);
M4=0;
else
if L<(L_Total-L5-L4) & L>=(L_Total-L5-L4-L3);
V0= W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2= W.*1000.*9.81.*cosd(Alpha);
V3=0;
V4=0;
M0=V0.*L;
M1=M0-(V0.*L1);
M2=M0-((L1+L2).*V0);
M3=0;
M4=0;
else
if L<(L_Total-L5-L4-L3) & L>=(L1);
V0= W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2=0;
V3=0;
V4=0;
M0=V0.*L;
M1=M0-(V0.*L1);
M2=0;
M3=0;
M4=0;
fprintf('V0=%d V1=%d V2=%d V3=%d V4=%d\n',V0,V1,V2,V3,V4);% in Newton
fprintf('M0=%d M1=%d M2=%d M3=%d M4=%d\n',M0,M1,M2,M3,M4);% in Newton-Meter
end
end
end
end
```

Comment: With this code, the internal forces and moments encountered by the boom can be found at any position and under load.

Input values

length, angle and weight. The number of booms opened according to the entered length and the internal forces and moments it encounters can be found.

Appendix

CALCULATE REACTION FORCE AND MOMENT FOR EACH BOOMS in MATLAB

```
>> L1=9;
L2=7.5;
L3=7;
L4=6.5;
L5=6;
L_Total=L1+L2+L3+L4+L5;
L=input('Enter Length of The Force Position in Meter\n');
W=input('Enter Value of W in Tonnes\n');
Alpha=input('Enter Value of Angle Alpha\n');
>> if L<=L_Total & L>=(L_Total-L5);
V0=W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2= W.*1000.*9.81.*cosd(Alpha);
V3= W.*1000.*9.81.*cosd(Alpha);
V4= W.*1000.*9.81.*cosd(Alpha);
M0=V0.*L;
M1=M0-(V0.*L1);
M2=M0-((L1+L2).*V0);
M3=M0-((L1+L2+L3).*V0);
M4=M0-((L1+L2+L3+L4).*V0);
else
if L<(L_Total-L5) & L>=(L_Total-L5-L4);
V0= W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2= W.*1000.*9.81.*cosd(Alpha);
V3= W.*1000.*9.81.*cosd(Alpha);
V4=0;
M0=V0.*L;
M1=M0-(V0.*L1);
M2=M0-((L1+L2).*V0);
M3=M0-((L1+L2+L3).*V0);
M4=0;
else
if L<(L_Total-L5-L4) & L>=(L_Total-L5-L4-L3);
V0= W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2= W.*1000.*9.81.*cosd(Alpha);
V3=0;
V4=0;
M0=V0.*L;
M1=M0-(V0.*L1);
M2=M0-((L1+L2).*V0);
M3=0;
M4=0;
else
if L<(L_Total-L5-L4-L3) & L>=(L1);
V0= W.*1000.*9.81.*cosd(Alpha);
V1= W.*1000.*9.81.*cosd(Alpha);
V2=0;
V3=0;
V4=0;
M0=V0.*L;
M1=M0-(V0.*L1);
M2=0;
M3=0;
M4=0;
fprintf('V0=%d V1=%d V2=%d V3=%d V4=%d\n',V0,V1,V2,V3,V4);% in Newton
fprintf('M0=%d M1=%d M2=%d M3=%d M4=%d\n',M0,M1,M2,M3,M4);% in Newton-Meter
end
end
```

```
end
end
```

CALCULATE FORCE, HYPOTENUSE and ANGLE in MATLAB

```
>> x=[9 6 5 12 8 5 16 14 10 5 18 16 10 5.4 22 20 18 14 10 5 26 24 22 20 14 10 5 30 28 26 22 18 14 10 6 32 28 24 18 14 10 6.3
36 34 30 26 22 18 14 8 38 36 32 28 22 18 14 8];
y=[0.3 5 5.3 2 7.3 9.2 1 6 10.2 13 5.8 9.2 14.3 16.4 5 9.4 12 15.8 18.4 20.2 3.3 9 12.2 14.8 20 22.2 23.2 1.1 8.3 12.8 17.8
21.3 24 26 27 7.7 15.8 20.7 25.2 28 29.2 30.2 5.3 11.8 19 23.2 27 29.4 31.3 33.2 7.6 12.3 20 24.8 30 32.5 34.2 36];
>> h=sqrt(x.^2+y.^2)
h =
Columns 1 through 12
    9.0050    7.8102    7.2863   12.1655   10.8301   10.4709   16.0312   15.2315   14.2843   13.9284   18.9114   18.4564
Columns 13 through 24
   17.4496   17.2662   22.5610   22.0989   21.6333   21.1102   20.9418   20.8096   26.2086   25.6320   25.1563   24.8805
Columns 25 through 36
   24.4131   24.3483   23.7327   30.0202   29.2043   28.9800   28.2991   27.8871   27.7849   27.8568   27.6586   32.9134
Columns 37 through 48
   32.1503   31.6937   30.9684   31.3050   30.8649   30.8501   36.3880   35.9894   35.5106   34.8459   34.8281   34.4726
Columns 49 through 58
   34.2883   34.1503   38.7525   38.0433   37.7359   37.4037   37.2022   37.1517   36.9546   36.8782
>> y./x
ans =
Columns 1 through 12
    0.0333    0.8333    1.0600    0.1667    0.9125    1.8400    0.0625    0.4286    1.0200    2.6000    0.3222    0.5750
Columns 13 through 24
    1.4300    3.0370    0.2273    0.4700    0.6667    1.1286    1.8400    4.0400    0.1269    0.3750    0.5545    0.7400
Columns 25 through 36
    1.4286    2.2200    4.6400    0.0367    0.2964    0.4923    0.8091    1.1833    1.7143    2.6000    4.5000    0.2406
Columns 37 through 48
    0.5643    0.8625    1.4000    2.0000    2.9200    4.7937    0.1472    0.3471    0.6333    0.8923    1.2273    1.6333
Columns 49 through 58
    2.2357    4.1500    0.2000    0.3417    0.6250    0.8857    1.3636    1.8056    2.4429    4.5000
>> atand(y./x)
ans =
Columns 1 through 12
    1.9092   39.8056   46.6683    9.4623   42.3805   61.4769    3.5763   23.1986   45.5673   68.9625   17.8601   29.8989
Columns 13 through 24
   55.0349   71.7749   12.8043   25.1735   33.6901   48.4566   61.4769   76.0973    7.2335   20.5560   29.0104   36.5014
Columns 25 through 36
   55.0080   65.7508   77.8378    2.0999   16.5113   26.2114   38.9760   49.7998   59.7436   68.9625   77.4712   13.5296
Columns 37 through 48
   29.4354   40.7778   54.4623   63.4349   71.0954   78.2166    8.3750   19.1398   32.3474   41.7428   50.8263   58.5232
Columns 49 through 58
   65.9018   76.4521   11.3099   18.8636   32.0054   41.5318   53.7462   61.0203   67.7379   77.4712
>> W=[26 43.6 55 14.8 29.1 42.3 9.1 11.4 21.2 42.2 7.7 9.1 21 38.2 5.4 6.4 7.6 11.6 19.7 30.6 4.1 4.7 5.5 6.5 11.5 18.4 22.8
3 3.4 3.9 5.4 7.5 10.9 16 18.8 2.5 3.3 4.5 7.5 10.6 13.8 14.7 1.7 2 2.7 3.6 5.1 7.3 9.4 11.8 1.4 1.7 2.3 3.1 5.1 7 8.4 10.1];
>> KG=W.*1000
KG =
Columns 1 through 10
   26000   43600   55000   14800   29100   42300   9100   11400   21200   42200
Columns 11 through 20
    7700    9100   21000   38200    5400    6400    7600   11600   19700   30600
Columns 21 through 30
    4100    4700    5500    6500   11500   18400   22800    3000    3400    3900
Columns 31 through 40
    5400    7500   10900   16000   18800    2500    3300    4500    7500   10600
Columns 41 through 50
   13800   14700   1700    2000    2700    3600    5100    7300    9400   11800
Columns 51 through 58
    1400    1700    2300    3100    5100    7000    8400   10100
```

```
>> F=KG.*9.81
```

```
F = (NEWTON)
```

```
Columns 1 through 10
```

```
255060 427716 539550 145188 285471 414963 89271 111834 207972 413982
```

```
Columns 11 through 20
```

```
75537 89271 206010 374742 52974 62784 74556 113796 193257 300186
```

```
Columns 21 through 30
```

```
40221 46107 53955 63765 112815 180504 223668 29430 33354 38259
```

```
Columns 31 through 40
```

```
52974 73575 106929 156960 184428 24525 32373 44145 73575 103986
```

```
Columns 41 through 50
```

```
135378 144207 16677 19620 26487 35316 50031 71613 92214 115758
```

```
Columns 51 through 58
```

```
13734 16677 22563 30411 50031 68670 82404 99081
```

```
F./1000= (KN)
```

```
Columns 1 through 12
```

```
255.0600 427.7160 539.5500 145.1880 285.4710 414.9630 89.2710 111.8340 207.9720 413.9820 75.5370 89.2710
```

```
Columns 13 through 24
```

```
206.0100 374.7420 52.9740 62.7840 74.5560 113.7960 193.2570 300.1860 40.2210 46.1070 53.9550 63.7650
```

```
Columns 25 through 36
```

```
112.8150 180.5040 223.6680 29.4300 33.3540 38.2590 52.9740 73.5750 106.9290 156.9600 184.4280 24.5250
```

```
Columns 37 through 48
```

```
32.3730 44.1450 73.5750 103.9860 135.3780 144.2070 16.6770 19.6200 26.4870 35.3160 50.0310 71.6130
```

```
Columns 49 through 58
```

```
92.2140 115.7580 13.7340 16.6770 22.5630 30.4110 50.0310 68.6700 82.4040 99.0810
```

```
>> x=[9 6 5 12 8 5 16 14 10 5 18 16 10 5.4 22 20 18 14 10 5 26 24 22 20 14 10 5 30 28 26 22 18 14 10 6 32 28 24 18 14 10 6.3  
36 34 30 26 22 18 14 8 38 36 32 28 22 18 14 8];
```

```
y=[0.3 5 5.3 2 7.3 9.2 1 6 10.2 13 5.8 9.2 14.3 16.4 5 9.4 12 15.8 18.4 20.2 3.3 9 12.2 14.8 20 22.2 23.2 1.1 8.3 12.8 17.8  
21.3 24 26 27 7.7 15.8 20.7 25.2 28 29.2 30.2 5.3 11.8 19 23.2 27 29.4 31.3 33.2 7.6 12.3 20 24.8 30 32.5 34.2 36];
```

```
>> z=atand(y./x)
```

```
z =
```

```
Columns 1 through 12
```

```
1.9092 39.8056 46.6683 9.4623 42.3805 61.4769 3.5763 23.1986 45.5673 68.9625 17.8601 29.8989
```

```
Columns 13 through 24
```

```
55.0349 71.7749 12.8043 25.1735 33.6901 48.4566 61.4769 76.0973 7.2335 20.5560 29.0104 36.5014
```

```
Columns 25 through 36
```

```
55.0080 65.7508 77.8378 2.0999 16.5113 26.2114 38.9760 49.7998 59.7436 68.9625 77.4712 13.5296
```

```
Columns 37 through 48
```

```
29.4354 40.7778 54.4623 63.4349 71.0954 78.2166 8.3750 19.1398 32.3474 41.7428 50.8263 58.5232
```

```
Columns 49 through 58
```

```
65.9018 76.4521 11.3099 18.8636 32.0054 41.5318 53.7462 61.0203 67.7379 77.4712
```

```
>> n=cosd(z)
```

```
n
```

```
Columns 1 through 12
```

```
0.9994 0.7682 0.6862 0.9864 0.7387 0.4775 0.9981 0.9191 0.7001 0.3590 0.9518 0.8669
```

```
Columns 13 through 24
```

```
0.5731 0.3128 0.9751 0.9050 0.8321 0.6632 0.4775 0.2403 0.9920 0.9363 0.8745 0.8038
```

```
Columns 25 through 36
```

```
0.5735 0.4107 0.2107 0.9993 0.9588 0.8972 0.7774 0.6455 0.5039 0.3590 0.2169 0.9722
```

```
Columns 37 through 48
```

```
0.8709 0.7572 0.5812 0.4472 0.3240 0.2042 0.9893 0.9447 0.8448 0.7461 0.6317 0.5222
```

```
Columns 49 through 58
```

```
0.4083 0.2343 0.9806 0.9463 0.8480 0.7486 0.5914 0.4845 0.3788 0.2169
```

```
>> F.*n
```

ans = The final state of the desired forces multiplied by the angle. The moment is ready to be taken just to be multiplied by the lengths. Columns 1 through 12

```
254.9184 328.5805 370.2502 143.2126 210.8732 198.1504 89.0972 102.7917 145.5953 148.6109 71.8967 77.3896
```

Columns 13 through 24

```
118.0597 117.2008 51.6567 56.8210 62.0343 75.4680 92.2828 72.1268 39.9009 43.1713 47.1854 51.2570
```

Columns 25 through 36

```
64.6952 74.1341 47.1224 29.4102 31.9786 34.3249 41.1825 47.4897 53.8784 56.3454 40.0081 23.8444
```

Columns 37 through 48

```
28.1940 33.4287 42.7646 46.5040 43.8615 29.4490 16.4992 18.5354 22.3767 26.3507 31.6032 37.3930
```

Columns 49 through 58

```
37.6512 27.1173 13.4673 15.7813 19.1334 22.7653 29.5865 33.2706 31.2182 21.4937
```

CALCULATE MOMENT in MATLAB

```
>> x=[9 6 5 12 8 5 16 14 10 5 18 16 10 5.4 22 20 18 14 10 5 26 24 22 20 14 10 5 30 28 26 22 18 14 10 6 32 28 24 18 14 10 6.3  
36 34 30 26 22 18 14 8 38 36 32 28 22 18 14 8];
```

```
y=[0.3 5 5.3 2 7.3 9.2 1 6 10.2 13 5.8 9.2 14.3 16.4 5 9.4 12 15.8 18.4 20.2 3.3 9 12.2 14.8 20 22.2 23.2 1.1 8.3 12.8 17.8  
21.3 24 26 27 7.7 15.8 20.7 25.2 28 29.2 30.2 5.3 11.8 19 23.2 27 29.4 31.3 33.2 7.6 12.3 20 24.8 30 32.5 34.2 36];
```

```
z=atand(y./x)
```

```
z =
```

Columns 1 through 12

```
1.9092 39.8056 46.6683 9.4623 42.3805 61.4769 3.5763 23.1986 45.5673 68.9625 17.8601 29.8989
```

Columns 13 through 24

```
55.0349 71.7749 12.8043 25.1735 33.6901 48.4566 61.4769 76.0973 7.2335 20.5560 29.0104 36.5014
```

Columns 25 through 36

```
55.0080 65.7508 77.8378 2.0999 16.5113 26.2114 38.9760 49.7998 59.7436 68.9625 77.4712 13.5296
```

Columns 37 through 48

```
29.4354 40.7778 54.4623 63.4349 71.0954 78.2166 8.3750 19.1398 32.3474 41.7428 50.8263 58.5232
```

Columns 49 through 58

```
65.9018 76.4521 11.3099 18.8636 32.0054 41.5318 53.7462 61.0203 67.7379 77.4712
```

```
>> n=cosd(z)
```

```
n =
```

Columns 1 through 12

```
0.9994 0.7682 0.6862 0.9864 0.7387 0.4775 0.9981 0.9191 0.7001 0.3590 0.9518 0.8669
```

Columns 13 through 24

```
0.5731 0.3128 0.9751 0.9050 0.8321 0.6632 0.4775 0.2403 0.9920 0.9363 0.8745 0.8038
```

Columns 25 through 36

```
0.5735 0.4107 0.2107 0.9993 0.9588 0.8972 0.7774 0.6455 0.5039 0.3590 0.2169 0.9722
```

Columns 37 through 48

```
0.8709 0.7572 0.5812 0.4472 0.3240 0.2042 0.9893 0.9447 0.8448 0.7461 0.6317 0.5222
```

Columns 49 through 58

```
0.4083 0.2343 0.9806 0.9463 0.8480 0.7486 0.5914 0.4845 0.3788 0.2169
```

```
>> W=[26 43.6 55 14.8 29.1 42.3 9.1 11.4 21.2 42.2 7.7 9.1 21 38.2 5.4 6.4 7.6 11.6 19.7 30.6 4.1 4.7 5.5 6.5 11.5 18.4 22.8  
3 3.4 3.9 5.4 7.5 10.9 16 18.8 2.5 3.3 4.5 7.5 10.6 13.8 14.7 1.7 2 2.7 3.6 5.1 7.3 9.4 11.8 1.4 1.7 2.3 3.1 5.1 7 8.4 10.1];
```

```
>> F=W.*9.81
```

```
F =
```

Columns 1 through 12

```
255.0600 427.7160 539.5500 145.1880 285.4710 414.9630 89.2710 111.8340 207.9720 413.9820 75.5370 89.2710
```

Columns 13 through 24

```
206.0100 374.7420 52.9740 62.7840 74.5560 113.7960 193.2570 300.1860 40.2210 46.1070 53.9550 63.7650
```

Columns 25 through 36

```
112.8150 180.5040 223.6680 29.4300 33.3540 38.2590 52.9740 73.5750 106.9290 156.9600 184.4280 24.5250
```

Columns 37 through 48

```
32.3730 44.1450 73.5750 103.9860 135.3780 144.2070 16.6770 19.6200 26.4870 35.3160 50.0310 71.6130
```

Columns 49 through 58

```
92.2140 115.7580 13.7340 16.6770 22.5630 30.4110 50.0310 68.6700 82.4040 99.0810
```

```

>> F.*n
ans =
Columns 1 through 12
254.9184 328.5805 370.2502 143.2126 210.8732 198.1504 89.0972 102.7917 145.5953 148.6109 71.8967 77.3896
Columns 13 through 24
118.0597 117.2008 51.6567 56.8210 62.0343 75.4680 92.2828 72.1268 39.9009 43.1713 47.1854 51.2570
Columns 25 through 36
64.6952 74.1341 47.1224 29.4102 31.9786 34.3249 41.1825 47.4897 53.8784 56.3454 40.0081 23.8444
Columns 37 through 48
28.1940 33.4287 42.7646 46.5040 43.8615 29.4490 16.4992 18.5354 22.3767 26.3507 31.6032 37.3930
Columns 49 through 58
37.6512 27.1173 13.4673 15.7813 19.1334 22.7653 29.5865 33.2706 31.2182 21.4937
>> m=F.*n
m =
Columns 1 through 12
254.9184 328.5805 370.2502 143.2126 210.8732 198.1504 89.0972 102.7917 145.5953 148.6109 71.8967 77.3896
Columns 13 through 24
118.0597 117.2008 51.6567 56.8210 62.0343 75.4680 92.2828 72.1268 39.9009 43.1713 47.1854 51.2570
Columns 25 through 36
64.6952 74.1341 47.1224 29.4102 31.9786 34.3249 41.1825 47.4897 53.8784 56.3454 40.0081 23.8444
Columns 37 through 48
28.1940 33.4287 42.7646 46.5040 43.8615 29.4490 16.4992 18.5354 22.3767 26.3507 31.6032 37.3930
Columns 49 through 58
37.6512 27.1173 13.4673 15.7813 19.1334 22.7653 29.5865 33.2706 31.2182 21.4937
>> x=[9 6 5 12 8 5 16 14 10 5 18 16 10 5.4 22 20 18 14 10 5 26 24 22 20 14 10 5 30 28 26 22 18 14 10 6 32 28 24 18 14 10 6.3
36 34 30 26 22 18 14 8 38 36 32 28 22 18 14 8];
y=[0.3 5 5.3 2 7.3 9.2 1 6 10.2 13 5.8 9.2 14.3 16.4 5 9.4 12 15.8 18.4 20.2 3.3 9 12.2 14.8 20 22.2 23.2 1.1 8.3 12.8 17.8
21.3 24 26 27 7.7 15.8 20.7 25.2 28 29.2 30.2 5.3 11.8 19 23.2 27 29.4 31.3 33.2 7.6 12.3 20 24.8 30 32.5 34.2 36];
h=sqrt(x.^2+y.^2)
h =
Columns 1 through 12
9.0050 7.8102 7.2863 12.1655 10.8301 10.4709 16.0312 15.2315 14.2843 13.9284 18.9114 18.4564
Columns 13 through 24
17.4496 17.2662 22.5610 22.0989 21.6333 21.1102 20.9418 20.8096 26.2086 25.6320 25.1563 24.8805
Columns 25 through 36
24.4131 24.3483 23.7327 30.0202 29.2043 28.9800 28.2991 27.8871 27.7849 27.8568 27.6586 32.9134
Columns 37 through 48
32.1503 31.6937 30.9684 31.3050 30.8649 30.8501 36.3880 35.9894 35.5106 34.8459 34.8281 34.4726
Columns 49 through 58
34.2883 34.1503 38.7525 38.0433 37.7359 37.4037 37.2022 37.1517 36.9546 36.8782
>> M=m.*h
M =
1.0e+03 *
Columns 1 through 12
2.2955 2.5663 2.6978 1.7423 2.2838 2.0748 1.4283 1.5657 2.0797 2.0699 1.3597 1.4283
Columns 13 through 24
2.0601 2.0236 1.1654 1.2557 1.3420 1.5931 1.9326 1.5009 1.0457 1.1066 1.1870 1.2753
Columns 25 through 36
1.5794 1.8050 1.1183 0.8829 0.9339 0.9947 1.1654 1.3243 1.4970 1.5696 1.1066 0.7848
Columns 37 through 48
0.9064 1.0595 1.3244 1.4558 1.3538 0.9085 0.6004 0.6671 0.7946 0.9182 1.1007 1.2890
Columns 49 through 58
1.2910 0.9261 0.5219 0.6004 0.7220 0.8515 1.1007 1.2361 1.1537 0.7926

```